

Iñigo Eguiluz Mateos

University of Deusto • Bilbao, España • eguiluz.inigom@gmail.com • +34 610 60 22 31
github.com/ineguiluz1 • [linkedin.com/in/iñigo-eguiluz-mateos](https://www.linkedin.com/in/iñigo-eguiluz-mateos) • ineguiluz.netlify.app

PROFESSIONAL SUMMARY

Data Science and Computer Engineering student specializing in **Quantitative Development and AI**. Technical expertise in high-performance computing (**C++/CUDA**), Reinforcement Learning applied to financial derivatives, and multimodal time-series forecasting. Strong mathematical foundation in stochastic processes and statistics.

EDUCATION

University of Deusto

Dual Bachelors Degree in Computer Engineering and Data Science & AI

Bilbao, España

Sept 2022 June 2027 (Expected)

GPA: 7.63 / 10.0

- **Relevant Coursework:** Calculus, Discrete Mathematics, Optimization, Machine Learning, Advanced Statistics, Algorithmics, and Big Data Processing, Reinforcement Learning, Natural Language Processing.
- **Academic Focus:** Specialized in the application of High-Performance Computing and Deep Learning to Financial Markets.

QUANTITATIVE & AI PROJECTS

High-Performance Option Pricing Engine

C++, CUDA, GPU Computing

- Developed a parallelized Monte Carlo simulation engine for option pricing, optimizing **CUDA kernel execution** and memory management to achieve massive parallel acceleration.
- Implemented stochastic volatility models to enhance pricing accuracy under complex market regimes.

Deep Hedging via Reinforcement Learning

Python, PyTorch, RL

- Engineered **SAC, TD3, and Recurrent TD3** agents to automate optimal option hedging under market frictions.
- Integrated **Behavioral Cloning** and **Curriculum Learning** to evolve strategies from classical Black-Scholes logic to robust, friction-aware policies.
- Validated on **real market data**, significantly reducing hedging error and tail-risk compared to traditional Delta-Hedging.

Multimodal Market Predictor

Amazon Chronos, FinBERT, Time-Series

- Developed a fusion architecture using Time-Series Foundation Models (**Amazon Chronos, PatchTST**) and **FinBERT semantic embeddings** from news to predict asset directionality.
- Framed the task as a tri-state classification (Long, Short, Neutral), integrating news context with temporal stock dynamics.

EXPERIENCE

DeustoTech

Research Intern (Computer Vision)

Bilbao, España

Noviembre 2025 Presente

- **High-Dimensional Data Processing:** Researching and developing 3D segmentation algorithms for large-scale **Point Clouds**, managing datasets with millions of spatial data points.
- **Performance Optimization:** Architecting efficient data ingestion pipelines, focusing on reducing memory overhead and improving computational throughput for real-time processing.
- **Feature Engineering:** Implementing spatial descriptors and geometric feature extraction methods to enhance model accuracy in unstructured 3D environments.

SKILLS & INTERESTS

- **Programming:** Python (Expert; NumPy, Pandas, PyTorch, ML Libraries), C++ (Advanced; CUDA), SQL, Linux Shell (Bash), Git, L^AT_EX.
- **Technical Expertise:** Parallel Computing, Deep Learning, Stochastic Processes, Time-Series Analysis, Reinforcement Learning.
- **Languages:** Spanish (Native), English (Advanced; CAE C1), Basque (Intermediate; B2), German (Basic).
- **Interests:** Quantitative Finance, Algorithmic Trading, Market Microstructure, High-Frequency Trading (HFT), Sport.